Oct. 10, 2022

Ellingsen + Palm 1975

inviscid, incompressible, unstratified flow, parallel flow $U(z)$, bounded by two horizontal planes

consider disturbance independent of x (though can still have u component)

total velocity $\underline{u}(y,z,t) = U(z)\,\underline{i} + \underline{u}'(y,z,t)$

momentum: $\frac{\partial \underline{u}}{\partial t} + \underline{u}\cdot\nabla\underline{u} = -\frac{1}{\rho_0}\nabla p + \nu\nabla^2\underline{u}$ (inviscid)

$p \neq p(x) \Rightarrow \frac{Du}{Dt} = 0$ in x

x-component of vorticity $\frac{D\zeta}{Dt} = 0$

$\frac{\partial u}{\partial x} = 0$ so continuity $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$ and can write $v = \frac{\partial \psi}{\partial z}$, $w = -\frac{\partial \psi}{\partial y}$

Linearizing: $\frac{\partial u}{\partial t} + wU' = 0$, $\frac{\partial}{\partial t}\nabla_H^2\psi = 0$ (2D Laplacian)

so $\nabla_H^2\psi$ not a function of time
so v, w not functions of time

$\Rightarrow \frac{\partial u}{\partial t} = -wU'$ RHS independent of time

$\Rightarrow \boxed{u = u(0) - wU't}$ u increases linearly with time

(paper also includes extensions to nonlinear case)

Stratification? same equation for x-mtm $\frac{Du}{Dt} = 0 \Rightarrow \frac{\partial u}{\partial t} + wU' = 0$ linear

vorticity: $\frac{\partial \underline{\omega}}{\partial t} + \underline{u}\cdot\nabla\underline{\omega} = \underline{\omega}\cdot\nabla\underline{u} + \frac{1}{\rho_0^2}\nabla\rho\times\nabla p$

x-component: $\frac{\partial \zeta}{\partial t} + \underline{u}\cdot\nabla\zeta = \underline{\omega}\cdot\nabla u + \frac{1}{\rho_0^2}\left(\frac{\partial \rho}{\partial y}\frac{\partial p}{\partial z} - \frac{\partial \rho}{\partial z}\frac{\partial p}{\partial y}\right)$ $\omega_1 = \zeta$

vortex stretching: $\omega_1 \cancel{\frac{\partial v}{\partial x}}^{0} + \omega_2 \frac{\partial u}{\partial y} + \omega_3 \frac{\partial u}{\partial z}$

$= \left(\frac{\partial u}{\partial z} - \cancel{\frac{\partial w}{\partial x}}^{0} \right) \frac{\partial u}{\partial y} + \left(\cancel{\frac{\partial v}{\partial x}}^{0} - \frac{\partial u}{\partial y} \right) \frac{\partial u}{\partial z} = 0$

so $\frac{\partial \zeta}{\partial t} + \underline{u} \cdot \nabla \zeta = \frac{1}{\rho_0^2} \left(\frac{\partial \rho}{\partial y} \frac{\partial p}{\partial z} - \frac{\partial \rho}{\partial z} \frac{\partial p}{\partial y} \right)$

p = deviation from hydrostatic

ρ = perturbation from background

$\Rightarrow$ baroclinic term is higher-order?

(I'm not sure if this is a bit dicey — would perturbations interact with background gradients? but see what we get for now)

linearizing (if we can safely drop the baroclinic term) gives

$\frac{\partial \zeta}{\partial t} = \frac{\partial}{\partial t} \nabla_H^2 \psi = 0$ as before

so $\nabla_H^2 \psi$ and therefore w are not functions of t, and

$\boxed{u = u(0) - \omega U' t}$ once again

this is sketchy! see next page

buoyancy: $\frac{\partial b}{\partial t} + \underline{u} \cdot \nabla b = \cancel{\kappa \nabla^2 b}^{\,=0}$ by assumption

linearized: $\frac{\partial b}{\partial t} + w N^2 = 0 \Rightarrow \boxed{b = b(0) - \omega N^2 t}$

how does this affect overall energy growth?

$E = \frac{1}{2} \langle \underline{u}, \underline{u} \rangle + \frac{Ri_b}{2} \langle b, b \rangle$ $\quad U, \omega$ independent of time

$= \frac{1}{2} \overline{(u(0) - \omega U' t)^2} + \frac{1}{2} \overline{v(0)^2} + \frac{1}{2} \overline{w(0)^2} + \frac{Ri_b}{2} \overline{(b(0) - \omega N^2 t)^2}$

$= \frac{1}{2} E(0) + \frac{1}{2} \overline{(\omega^2 U'^2 t^2 - 2u(0) \omega U' t)} + \frac{Ri_b}{2} \overline{(\omega^2 N^4 t^2 - 2b(0) \omega N^2 t)}$

Q: do background shear and background stratification have same effect on perturbation energy here?

try $\psi = A \sin k_1 y \sin k_2 z \Rightarrow v = \frac{\partial \psi}{\partial z} = k_2 A \sin k_1 y \cos k_2 z$, $\quad u(0) = U(z)$, $b(0) = B(z)$

$w = -\frac{\partial \psi}{\partial y} = -k_1 A \cos k_1 y \sin k_2 z$,

but wait – how is w affected by stratification?

argument for $w \neq w(t)$ is based on continuity and vorticity – does this still hold?

$$\frac{\partial w}{\partial t} + \underline{u} \cdot \nabla w = -\frac{1}{\rho_0}\frac{\partial p}{\partial z} + b$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho_0}\frac{\partial p}{\partial z} + b$$

linearized: $\frac{\partial w}{\partial t} = -\frac{1}{\rho_0}\frac{\partial p}{\partial z} + b$ but $b = b(t)$ now! so probably can't drop baroclinic terms, above

$$\frac{\partial v}{\partial t} + \underline{u} \cdot \nabla v = -\frac{1}{\rho_0}\frac{\partial p}{\partial y}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = -\frac{1}{\rho_0}\frac{\partial p}{\partial y}$$

linearized: $\frac{\partial v}{\partial t} = -\frac{1}{\rho_0}\frac{\partial p}{\partial y}$

$$\Rightarrow \frac{\partial}{\partial t}\left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}\right) = \boxed{\frac{\partial \zeta}{\partial t} = -\frac{\partial b}{\partial y}}$$

means we can't argue $w \neq w(t)$ anymore

what can we say?

have: $\frac{\partial u}{\partial t} + wU' = 0$ + continuity $\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$

$\frac{\partial b}{\partial t} + wN^2 = 0$

$\frac{\partial \zeta}{\partial t} + \frac{\partial b}{\partial y} = 0$

write in terms of streamfunction:

$$\left.\begin{aligned} &\frac{\partial u}{\partial t} - \frac{\partial \psi}{\partial y}U' = 0 \\ &\frac{\partial b}{\partial t} - \frac{\partial \psi}{\partial y}N^2 = 0 \\ &\frac{\partial \nabla_H^2 \psi}{\partial t} + \frac{\partial b}{\partial y} = 0 \end{aligned}\right\}$$

decouples

2 equations, 2 unknowns

for case with N^2 = constant (uniform background shear):

$$\left.\begin{aligned} \frac{\partial b}{\partial t} - N^2 \frac{\partial \psi}{\partial y} &= 0 \\ \frac{\partial \nabla_H^2 \psi}{\partial t} + \frac{\partial b}{\partial y} &= 0 \end{aligned}\right\} \quad \frac{\partial^2 \nabla_H^2 \psi}{\partial t^2} + N^2 \frac{\partial^2 \psi}{\partial y^2} = 0$$

assume sinusoidal structure in y, z for ψ, but keep time dependence generic:

$$\psi = A e^{ik_1 y} e^{ik_2 z} f(t)$$

$$-A(k_1^2 + k_2^2) e^{ik_1 y} e^{ik_2 z} \frac{d^2 f}{dt^2} - N^2 A k_1^2 e^{ik_1 y} e^{ik_2 z} f = 0 \quad \leftarrow \text{gives exponential time dependence}$$

$$(k_1^2 + k_2^2) \frac{d^2 f}{dt^2} - N^2 k_1^2 f = 0$$

$$\frac{d^2 f}{dt^2} = \frac{N^2 k_1^2}{k_1^2 + k_2^2} f \qquad \text{internal waves?}$$

ψ without b terms, get $\frac{d^2 f}{dt^2} = 0 \Rightarrow f = At^2 + Bt + C$

→ wavelike behaviour for spanwise-2D perturbations
technically no preferred direction, but shear + Orr set this in xz plane (rather than yz-plane or oblique?)

come back to this later...

MHD case?

linearized equations used in Dedalus (from overleaf): (note nondimensional)

$$\left.\begin{aligned} \frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + wU' &= -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u \\ \frac{\partial v}{\partial t} + U\frac{\partial v}{\partial x} &= -\frac{\partial p}{\partial y} + \frac{j_z}{M_A^2} + \frac{1}{Re}\nabla^2 v \\ \frac{\partial w}{\partial t} + U\frac{\partial w}{\partial x} &= -\frac{\partial p}{\partial z} - \frac{j_y}{M_A^2} + \frac{1}{Re}\nabla^2 w \\ \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} &= 0 \end{aligned}\right\} \text{ momentum}$$

$$\frac{\partial A_x}{\partial t} + \frac{\partial \Phi}{\partial x} - v = \frac{1}{R_m}\nabla^2 A_x$$
$$\frac{\partial A_y}{\partial t} + \frac{\partial \Phi}{\partial y} - w - Ub_x = \frac{1}{R_m}\nabla^2 A_y$$
$$\frac{\partial A_z}{\partial t} + \frac{\partial \Phi}{\partial z} - Ub_y = \frac{1}{R_m}\nabla^2 A_z$$
$$\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0$$

$\underline{B} = \nabla \times \underline{A}$

$\underline{J} = \nabla \times \underline{B} = 0$

as before, simplify: no diffusive terms, no x-derivatives

$$\Rightarrow \frac{\partial u}{\partial t} + wV' = 0 \qquad \frac{\partial A_x}{\partial t} - v = 0$$
$$\frac{\partial v}{\partial t} = -\frac{\partial p}{\partial y} + \frac{j_z}{M_A^2} \qquad \frac{\partial A_y}{\partial t} - w + \frac{\partial \Phi}{\partial y} = 0$$
$$\frac{\partial w}{\partial t} = -\frac{\partial p}{\partial z} - \frac{j_y}{M_A^2} \qquad \frac{\partial A_z}{\partial t} - Ub_y + \frac{\partial \Phi}{\partial z} = 0$$
$$\frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \qquad \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0$$

vorticity: $\frac{\partial}{\partial t}\left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}\right) = \frac{\partial \zeta}{\partial t} = \frac{1}{M_A^2}\left(\frac{\partial j_z}{\partial z} + \frac{\partial j_y}{\partial y}\right)$

but $j_z = \frac{\partial b_y}{\partial x}^{=0} - \frac{\partial b_x}{\partial y} = -\frac{\partial b_x}{\partial y}$

$j_y = \frac{\partial b_x}{\partial z} - \frac{\partial b_z}{\partial x}^{=0} = \frac{\partial b_x}{\partial z}$

so $\frac{\partial \zeta}{\partial t} = \frac{1}{M_A^2}\left(\frac{\partial}{\partial z}\left(-\frac{\partial b_x}{\partial y}\right) + \frac{\partial}{\partial y}\left(\frac{\partial b_x}{\partial z}\right)\right) = 0$

in this case, $\frac{\partial \zeta}{\partial t} = \frac{\partial \nabla^2 \psi}{\partial t} = 0 \Rightarrow \nabla^2\psi$ independent of t

$\Rightarrow u, w$ independent of t (as in original Ellingsen + Palm paper)

$$\Rightarrow \boxed{u = u(0) - wV't}$$

if v = constant wrt time, then $\frac{\partial A_x}{\partial t} = v \Rightarrow \boxed{A_x = A_x(0) + vt}$

x-component of B: $B_x = \frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y} \Rightarrow \frac{\partial B_x}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y}\right) = \frac{\partial w}{\partial z} + U\frac{\partial b_y}{\partial y}$

SUMMARY THUS FAR

For inviscid incompressible parallel shear flows $U(z)$ (in x-direction), what happens when we perturb flow in yz-plane ($\partial/\partial x = 0$)?

Unstratified: $\frac{\partial u}{\partial t} + wU' = 0$

$\frac{\partial \zeta}{\partial t} = \frac{\partial \nabla_H^2 \psi}{\partial t} = 0$ for $v = \frac{\partial \psi}{\partial z}$, $w = -\frac{\partial \psi}{\partial y}$ $\Rightarrow$ v, w independent of t

ζ = x-component of vorticity

$\boxed{u = u(0) - wU't}$

u grows linearly with structure in yz-plane

Stratified: $N^2 = 0$ background

$\frac{\partial u}{\partial t} + wU' = 0$

$\frac{\partial b}{\partial t} + wN^2 = \frac{\partial b}{\partial t} - N^2 \frac{\partial \psi}{\partial y} = 0$

$\frac{\partial \zeta}{\partial t} = \frac{\partial \nabla_H^2 \psi}{\partial t} = -\frac{\partial b}{\partial y}$; separable solutions suggest internal waves in yz plane

lift-up not giving algebraic growth

suggests internal wave behaviour for u as well

no reason for it to depend on direction, but Orr in xz-plane gives sort of "boost" to waves, which may be enough to make this behaviour optimal

MHD: velocities give $\frac{\partial u}{\partial t} + wU' = 0$ and $\frac{\partial \zeta}{\partial t} = \frac{\partial \nabla_H^2 \psi}{\partial t} = 0$ for spanwise perturbations $\rightarrow$ same as unstratified liftup!

suggests velocities v, w independent of time, $u = u(0) - wU't$ Lift-up

haven't yet figured out impact on magnetic field, though

suggests fundamental difference with stratified case: stratification completely alters long-time lift-up behaviour, but magnetic fields still permit this (at least for the velocity)

$\rightarrow$ at longer target times, if this is correct it would mean algebraic growth from spanwise-2D perturbations could be optimal, at least until magnetic fields alter this or a normal mode instability takes over